

Deciphering Gödel

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Abstract. The Gödel papers contain several thousand pages of notes on results in logic and foundations. The main part of these, some 1700 pages, were written in 1940-42, but their contents have been hidden behind the veil of an archaic, forgotten German shorthand. After a description of the shorthand, its deciphering, and the nature of the sources, three examples of results will be discussed. Two of them are cleared by methods somewhat different from those of Gödel. The third one is a claim shown to be false by a counterexample, the correction of which leads to a surprising new result on the relation between classical and intuitionistic logic.

Part I: Principles of shorthand writing

-A *phonetic notation* designed to be clearly faster than ordinary speech

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- Machine reading harder by orders of magnitude than standard handwriting (courtesy Chrome AI mode):

The primary paradox in handwriting recognition is known as Sayre's Paradox, which dictates that a handwritten cursive word cannot be recognized without being segmented into individual letters, yet it cannot be correctly segmented into letters until the whole word is recognized.

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- Reading of shorthand is in part an analytic process of figuring out the consonants, in part a synthetic view of a sentence and the words in it as a whole

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t									
v									
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z									

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- leucht	Re	- lassen	Do	D
- eifern	Dn	- lässigkeit	Do	D
- eilen	Dn	- laufen	Dn	D
- einander	De	- liefern	Dn	D
- empfinden	Dn	- machen	Do	D
Nachen ¹¹	D	- mals	Re	
Nacherte	Dn	- meinem Dapier ²²		
- erzählung	Dn	- halten	Dn	D
- fahren	Dn	- mittag ^{34, 35}	Do	D
- feier	Dn	- nahme	Dn	D
- folger	Dn	Nachod	Re	
- forschung	Dn	nachspießen ²³	Dn	D
- frage	Dn	- plaudern	Dn	
- fühlen	Dn	- prägen	Dn	
- geben	Dn	- rede	Re	D
- geburt	Dn	- richt	Do	D
- gerade	Dn	- ruf	Dn	D
- giebig	Dn	- ruhm	Re	D
- haltig	Dn	- sagen	Dn	Do
- her	Re	- sate	Dn	D
- hilfe	Dn	- schicken	Dn	D
- hut	Dn	- schlagen	Dn	D
- jagen	Dn	- schlüssel	Do	D
- klingen	Dn	- schreiben	Dn	D

Nachsicht	Dn	Nachtisch	Dn	D
- silbe	Re	- wache	Re	D
- sitzen	Dn	- zeit	Dn	D
- sommer	Dn	Nachübung	Dn	D
- spiel	Dn	- wachsen	Re	D
- sprechen	Dn	- weis	Re	D
nüchot ¹¹	D	- welt	Re	D
Nachstellung	Dn	- wort	Re	D
Nachstentliebe	Dn	- wuchs	Dn	D
nächstens	Dn	- zahlen	Dn	D
Nacht	Dn	- zug	Dn	D
Nachtteil	Dn	- zügle	Dn	D
Nachtgall	Dn	Nachen		
Nachtigall	Dn	Nachtheit		
nächtigen	Dn	Nadel	Re	
Nachtisch	Dn	Nagel	Re	
Nachtlager	Dn	nagen	Dn	
nächtlich	Dn	Nager	Dn	
Nachtmahl	Dn	nahe	Dn	
- quartier	Dn	Nähe	Dn	
Nachttrab	Dn	- en	Re	
- trüglich ²³	Dn	- er	Re	
Nachtregen	Dn	- erin	Re	
nachts	Dn	- maschine	Dn	Do
Nachtschatten	Dn	- nadel ³⁴	Re	Do

Problems with Gödel

- undisciplined writing
- often just formulas page after page
- at its worst: horrendous pages with cancellations and additions
- at its best: orderly writing with practically no changes

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Jan von Plato

Chapters from Gödel's Unfinished Book on Foundational Research in Mathematics

This volume contains English translations of Gödel's chapters on logicism and the antinomies and on the calculi of pure logic, as well as outlines for a chapter on metamathematics. It also comprises most of his reading notes.

This book is a testimony to Gödel's understanding of the situation of foundational research in mathematics after his great discovery, the incompleteness theorem of 1931. It is also a source for his views on his logical predecessors, from Leibniz, Frege, and Russell to his own times. Gödel's 'own book on foundations,' as he called it, is essential reading for logicians and philosophers interested in foundations. Furthermore, it opens a new chapter to the life and achievement of one of the icons of 20th century science and philosophy.

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Sources and Studies in the History of Mathematics
and Physical Sciences

Jan von Plato

Can Mathematics Be Proved Consistent?

Gödel's Shorthand Notes &
Lectures on Incompleteness

 Springer

1./II. 1941

Jede Fmct d. eigentl. intmit. Systems ist berechenbarZf. $e \in \mathcal{P}(\mathcal{N})$ p. 6' e. s. Ark.-H. 7. p. 34 $[\mathcal{P}(\mathcal{P}(\mathcal{N})) \mathcal{P}(\mathcal{P}(\mathcal{N}))]$ Zf. $\sim \mathcal{P}(\mathcal{N})^{\mathcal{N}}$ p. 0 ~ 20 Wk $\sim \sim \mathcal{P}(\mathcal{P}(\mathcal{N}))$ m. v. d. e.
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Th. 48 $\sim \mathcal{L}(\mathcal{U}_1 \dots \mathcal{U}_n \text{ Wk}^{\mathcal{N}} \text{ u. } \mathcal{U}_1 \dots \mathcal{U}_n \sim$ $\mathcal{P}(\mathcal{N})/\mathcal{N} \text{ u. d. } \mathcal{L}(\mathcal{U}_1 \dots \mathcal{U}_n) \text{ Wk}^{\mathcal{N}}$ Bew. $\sim \mathcal{P}(\mathcal{N})/\mathcal{N} \text{ u. d. } \mathcal{L}(\mathcal{U}_1 \dots \mathcal{U}_n) \text{ Wk}^{\mathcal{N}}$ $\mathcal{P}(\mathcal{P}(\mathcal{N})/\mathcal{N}) \mathcal{U}_1 \dots \mathcal{U}_n \text{ Wk}^{\mathcal{N}}, \mathcal{L}(\mathcal{U}_1 \dots \mathcal{U}_n) \sim$
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1./I. 1941

Each function of the properly intuitionistic system is computable

Definition. The formal system laid as a basis is the one of *Arbeitsheft* 7, reverse direction page 34 [with a fixed numbering of the defined constants].

Definition. A constant expression²⁰ $\mathfrak{A}$ of level 0 is called computable if there exists a numeral n such that

$\mathfrak{A} = n$ is provable

A constant expression $\mathfrak{B}$ of level $\mathfrak{S}_{k+1}$ is called computable if there exists, for each (type-wise) suitable complete argument series of constant expressions $\mathfrak{A}_1 \dots \mathfrak{A}_r$, a numeral n such that

$\mathfrak{B}(\mathfrak{A}_1 \dots \mathfrak{A}_r) = n$ is provable²¹

(The expressions $\mathfrak{A}_1 \dots \mathfrak{A}_r$ belong to the sets $\mathfrak{S}_k + \dots + \mathfrak{S}_0$.)

Theorem 48. If $\mathfrak{B}$ and $\mathfrak{A}_1 \dots \mathfrak{A}_n$ are computable and $\mathfrak{A}_1 \dots \mathfrak{A}_n$ a suitable argument series, then also $\mathfrak{B}(\mathfrak{A}_1 \dots \mathfrak{A}_n)$ is computable.

Proof. Because each suitable complete argument series of computable constant expressions for this expression gives together with $\mathfrak{A}_1 \dots \mathfrak{A}_n$ a complete suitable argument series of computable constant expressions for $\mathfrak{B}$. Therefore there exists a numeral n such that the equation is provable.

Sources and Studies in the History of Mathematics and Physical Sciences

Maria Hämeen-Anttila · Jan von Plato · Editors

Kurt Gödel: Results on Foundations

Kurt Gödel (1906–1978) gained world-wide fame by his incompleteness theorem of 1931. Later, he set as his aim to solve what are known as Hilbert's first and second problems, namely Cantor's continuum hypothesis about the cardinality of real numbers, and secondly the consistency of the theory of real numbers and functions. By 1940, he was halfway through the first problem, in what was his last published result in logic and foundations. His intense attempts thereafter at solving these two problems have remained behind the veil of a forgotten German shorthand he used in all of his writing.

Results on Foundations is a set of four shorthand notebooks written in 1940–42 that collect results Gödel considered finished. Its main topic is set theory in which Gödel anticipated several decades of development. Secondly, Gödel completed his 1933 program of establishing the connections between intuitionistic and modal logic, by methods and results that today are at the same time new and 80 years old.

The present edition of Gödel's four notebooks encompasses the 368 numbered pages and 126 numbered theorems of the *Results on Foundations*, together with a list of 74 problems on set theory Gödel prepared in 1946, and a list of an unknown date titled "The grand program of my research in ca. hundred questions."

ISBN 978-3-031-37874-4



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Kurt Gödel: Results on Foundations

Sources and Studies in the History of Mathematics
and Physical Sciences

Maria Hämeen-Anttila
Jan von Plato *Editors*

Kurt Gödel: Results on Foundations

 Springer

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Vorlesung Sommer
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[illegible]

P. voss & co. h. u.

$$c \geq \frac{1}{4n} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$$

for 100 Hz $\frac{d\phi}{dt} = 2\pi \times 100 \times \frac{1}{4} = 157 \text{ rad/sec}$

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Def 4. A graph G is a $\frac{p}{q}$ -graph if and only if

$\angle$ side SP up $\angle$ side up $\frac{y}{2}$.

Part I: Grundzüge des Logik-Kalküls

Hs 973:218

Vorlesung Sommer 1934

*Erste Vorlesung*¹

Wenn wir schnell in die mathematische Logik und ihre Beziehung zur Mathematik hineinkommen wollen, so empfiehlt es sich, dass wir uns an diejenige Art der Anwendung logischer Formeln halten, auf welche die mathematische Logik in ihrer früheren Form sozusagen zugeschnitten ist. Das ist die axiomatische Form [added: oder Axiomatik] der Behandlung einer Theorie. [Written above and between the lines of the last sentence: Diese ist am ausgeprägtesten in der Form, die sie durch Weierstrass Cantor Dedekind erhalten hat, Analysis, die Mengenlehre, die moderne Algebra.]

Wir haben da eine Gattung oder mehrere von Dingen (Individuen). Zu jede Gattung gehört ein *Individuen-Bereich*, ferner gewisse Grundprädikate und Grund-Beziehungen in Hinsicht auf der die Dinge betrachtet werden.

Es werden gewisse Aussagen über diese Grundprädikate (Grundbeziehungen) als Axiome vorangestellt und aus diesen andere logisch gefolgert, das geht so, dass dabei nichts anderes benutzt wird als eben die voran gestellten Axiome.

Das klassische Beispiel einer in dieser Weise ausgebildeten Theorie ist die Geometrie, und zwar können wir uns hier auf die Geometrie der Ebene beschränken und brauchen auch von dieser nur diejenigen Beziehungen zu betrachten, welche die Lageverhältnisse von Punkte und Geraden und die Anordnung von Punkte auf den Geraden betreffen.

2

Ehe wir uns dieses genauer vornehmen sollen, seien zunächst ein Paar Vorbemerkungen angebracht.

1. Historisch: Die Art der Axiomatik, wie wir sie betrachten, stimmt nicht mit derjenigen bei Euklid überein, und zwar aus zwei ganz

¹ Original page numbering begins with the second lecture.

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WA 77

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Formulas: Sütterlin to Fraktur

9

(X)

$$\frac{\frac{\alpha}{\alpha \& \mathcal{L}}}{\alpha} \begin{matrix} \beta \\ \mathcal{L} \end{matrix} \begin{matrix} VE \\ VB \end{matrix} \text{ mind zu: } \begin{matrix} \alpha \\ E \end{matrix}$$

(V)

$$\frac{\frac{\alpha}{\alpha \vee \mathcal{L}}}{\mathcal{L}} \begin{matrix} [\alpha] \\ \gamma \end{matrix} \begin{matrix} [\mathcal{L}] \\ \delta \end{matrix} \begin{matrix} OE \\ OB \end{matrix} \text{ zu: } \begin{bmatrix} \alpha \\ \sigma \end{bmatrix} \begin{matrix} \gamma \\ \mathcal{L} \\ E \end{matrix}$$

(Bei der anderen Form von VB bzw. OE ist abgang analog.)

(.)

$$\frac{\frac{\alpha(a)}{\mathcal{P}a}}{\mathcal{P}b} \begin{matrix} AE \\ AB \end{matrix} \text{ zu: } \begin{matrix} \alpha(b) \\ \mathcal{P}b \\ E \end{matrix}$$

(E)

$$\frac{\frac{\alpha}{\mathcal{P}a}}{\mathcal{L}} \begin{matrix} [\mathcal{P}b] \\ \gamma(b) \end{matrix} \begin{matrix} SE \\ SB \end{matrix} \text{ zu: } \begin{bmatrix} \alpha \\ \mathcal{P}a \end{bmatrix} \begin{matrix} \gamma(a) \\ \mathcal{L} \\ E \end{matrix}$$

(→)

$$\frac{\frac{\alpha}{\alpha} \frac{\gamma}{\mathcal{L}}}{\mathcal{L}} \begin{matrix} [\alpha] \\ \gamma \end{matrix} \begin{matrix} FE \\ FB \end{matrix}$$

zu:

$$\begin{bmatrix} \alpha \\ \sigma \end{bmatrix} \begin{matrix} \gamma \\ \mathcal{L} \\ E \end{matrix}$$

Es bedarf einiger
Übungen, um
nanzufassen, daß in der
Art jenseits werden ein
wichtiges Linsen nutzt.



The Sütterlin alphabet was still taught at the *Deutsche Schule Helsinki* around 1960. This illustration with the title *Buchstabenbilder* and beautiful handwritten Sütterlin letters is from the author's personal copy of the book of readings *Lebendige Sprache. Ein Wegweiser in die Muttersprache* (part 2 for the third and fourth school-years), written by Hans Schorer and Albert Wiechmann and published by Verlag Moritz Diesterweg (tenth printing, 1959).

Formulas: Sütterlin to Fraktur

ULS 9

$$\begin{array}{ccc}
 \& & \vee \\
 \frac{\frac{\alpha}{\mathfrak{A}} \quad \frac{\beta}{\mathfrak{B}}}{\mathfrak{A} \& \mathfrak{B}} \begin{array}{l} AI \\ AE \end{array} & \text{becomes:} & \frac{\frac{\alpha}{\mathfrak{A}}}{\mathfrak{A}} \begin{array}{l} \\ \varepsilon \end{array} \\
 & & \frac{\frac{\alpha}{\mathfrak{A}} \quad \frac{[\mathfrak{A}]}{\gamma} \quad \frac{[\mathfrak{B}]}{\delta}}{\mathfrak{A} \vee \mathfrak{B}} \begin{array}{l} OI \\ \mathfrak{C} \end{array} \quad \frac{\mathfrak{C}}{\mathfrak{C}} \begin{array}{l} OE \\ \varepsilon \end{array} \quad \text{into:} \quad \frac{[\frac{\alpha}{\mathfrak{A}}]}{\gamma} \begin{array}{l} \\ \mathfrak{C} \\ \varepsilon \end{array}
 \end{array}$$

(it is quite analogous with the other form of AE resp. OI .)

()

(E)

$$\begin{array}{ccc}
 \frac{\frac{\alpha(a)}{\mathfrak{P}a}}{(\mathfrak{r})\mathfrak{P}\mathfrak{r}} \begin{array}{l} UI \\ UE \end{array} & \text{into:} & \frac{\frac{\alpha(b)}{\mathfrak{P}b}}{\mathfrak{P}b} \begin{array}{l} \\ \varepsilon \end{array} \\
 & & \frac{\frac{\alpha}{\mathfrak{P}a} \quad \frac{[\mathfrak{P}b]}{\gamma(b)}}{(\mathfrak{E}\mathfrak{r})\mathfrak{P}\mathfrak{r}} \begin{array}{l} EI \\ \mathfrak{C} \end{array} \quad \frac{\mathfrak{C}}{\mathfrak{C}} \begin{array}{l} EE \\ \varepsilon \end{array} \quad \text{into:} \quad \frac{[\frac{\alpha}{\mathfrak{P}a}]}{\gamma(a)} \begin{array}{l} \\ \mathfrak{C} \\ \varepsilon \end{array}
 \end{array}$$

→

$$\begin{array}{ccc}
 \frac{\frac{\alpha}{\mathfrak{A}} \quad \frac{[\mathfrak{A}]}{\gamma} \quad \frac{[\mathfrak{B}]}{\delta}}{\mathfrak{A} \rightarrow \mathfrak{B}} \begin{array}{l} FI \\ FE \end{array} & \text{into:} & \frac{[\frac{\alpha}{\mathfrak{A}}]}{\gamma} \begin{array}{l} \\ \mathfrak{B} \\ \varepsilon \end{array}
 \end{array}$$

It requires some considerations to see to it that a correct proof is in fact produced in each case. I shall refrain from the exact realization, because I will not make any use of these facts, but rather present them for the sake of intuitiveness. —

Sources and Studies in the History of Mathematics and Physical Sciences

Jan von Plato

Saved from the Cellar

Gerhard Gentzen's Shorthand Notes on Logic and Foundations of Mathematics

Gerhard Gentzen is best known for his development of the proof systems of natural deduction and sequent calculus, central in many areas of logic and computer science today. Another noteworthy achievement is his resolution of the embarrassing situation created by Gödel's incompleteness results, especially the second one about the unprovability of consistency of elementary arithmetic. After these successes, Gentzen dedicated the rest of his short life to the main problem of Hilbert's proof theory, the question of the consistency of analysis. He was arrested in the summer of 1945 with other professors of the German University of Prague and died soon afterward of starvation in a prison cell. Attempts at locating his lost manuscripts failed at the time, but several decades later, two slim folders of shorthand notes were found. In this volume, Jan von Plato gives an overview of Gentzen's life and scientific achievements, based on detailed archival and systematic studies, and essential for placing the translations of shorthand manuscripts that follow in the right setting. The materials in this book are singular in the way they show the birth and development of Gentzen's central ideas and results, sometimes in a well-developed form, and other times as flashes into the anatomy of the workings of a unique mind.

Mathematics

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Sources and Studies in the History of Mathematics
and Physical Sciences

Jan von Plato

Saved from the Cellar

Gerhard Gentzen's Shorthand Notes on
Logic and Foundations of Mathematics

 Springer

PART II: The texts deciphered, what then?

Three examples:

1. A decision method for positive logic
2. The modal interpretation of intuitionistic logic
3. "The constructively strongest formula"

1. Positive propositional logic

-Separate out the logical principles about implication, an idea of Paul Bernays

-The axioms of implication found already in Frege's *Begriffsschrift*:

1. $A \supset (B \supset A)$
2. $(A \supset (A \supset B)) \supset (A \supset B)$
3. $(A \supset (B \supset C)) \supset (B \supset (A \supset C))$
4. $(A \supset B) \supset ((B \supset C) \supset (A \supset C))$

1. Positive propositional logic

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$$2. (A \supset (A \supset B)) \supset (A \supset B)$$

$$3. (A \supset (B \supset C)) \supset (B \supset (A \supset C))$$

$$4. (A \supset B) \supset ((B \supset C) \supset (A \supset C))$$

-We can add negation to the language if we define:

$$\sim A \equiv A \supset \perp$$

Hertz systems

- By Paul Hertz, Göttingen mathematician, 1922
- Gentzen's first work about Hertz systems, 1932, introduced the notion of *Cut*:

$$\frac{\Gamma \rightarrow b \quad b, \Delta \rightarrow c}{\Gamma, \Delta \rightarrow c} \textit{Cut}$$

$$\mu \quad \text{Th 3} \quad R(xyz) \equiv$$

$$(\exists x_1 \dots x_n)(y_1 \dots y_n) f(xyz x_1 \dots x_n y_1 \dots y_n) \neq 0$$

$$\text{ob } (\exists x) \varphi(x) \equiv (\exists x y x_1 \dots x_n)(y_1 \dots y_n)$$

$$[f(x y \underline{m} x_1 \dots x_n y_1 \dots y_n) \neq 0]$$

Positive Aussagenkalkül. *

σ - vor $\wedge$ & $\vee$ $a, b, \dots$

Df. Aussage für σ $\sim$ $\text{keine } a, b, \dots \rightarrow c$

$\wedge$ & $\vee$ per $\sim$ & $\sim$

Df. System von Aussagen (M), σ

* vgl. lose Blätter 1-11 u. 25. Positive Aussagenkalkül

$$2 - \int \mathcal{R} \mapsto n \leq \mathcal{L} \mathcal{A} a$$

$$1. \mapsto \mathcal{A} a \mapsto a \quad a \mapsto \mathcal{R} \mapsto$$

$$2. \mapsto \mathcal{R} \mapsto \mathcal{R} \mapsto \mathcal{A} \mapsto a : \mathcal{A}, b \mapsto a \quad *$$

$$3. \quad " \quad " \quad " \quad " \quad \mathcal{A} \mapsto a \quad \mathcal{L}, a \mapsto b \quad \left. \vphantom{\begin{matrix} \mathcal{A} \mapsto a \\ \mathcal{L}, a \mapsto b \end{matrix}} \right\} \mathcal{A}, \mathcal{L} \mapsto b$$

Df. Exportationsregel f. a, b, c ($\mathcal{R}_{abc}$)

$$\mapsto \mathcal{R} \mapsto (\mathcal{A} \mapsto \mathcal{L} \mathcal{A})$$

$$\mathcal{A}, a \mapsto b : \mathcal{A} \mapsto c$$

$$* (\mapsto \mathcal{R} \mapsto \mathcal{L} \mathcal{A} \quad c = a \mapsto b)$$

Df. $m \sim \mathcal{R} \quad \{ a, b, c \in \mathcal{R} \quad \mathcal{R}$

$$m \sim \mathcal{R} \quad \{ a, b, c \in \mathcal{R} \quad \mathcal{R}$$

Gödel's work on positive propositional logic:

- Resultate Grundlagen* pp. 66–78

- 11-page addition of proofs to previous

- the *Logik und Grundlagen* series: pp. 295–302, 333–334, 336–339, 340–348, 361–362

Most of this was written soon after Gödel arrived in Princeton for good, March 1940

System NDL

-From the Notre Dame Lectures on Logic:

1. *Initial sequents*
2. *Weakening*
3. *Right implication*

$$\frac{A, \Gamma \rightarrow B}{\Gamma \rightarrow A \supset B} R\supset$$

4. *Modus Ponens:*

$$\frac{\Gamma \rightarrow A \supset B \quad \Gamma \rightarrow A}{\Gamma \rightarrow B} MP$$

Note that rule MP has *shared contexts*. Without them, we would have as in Gentzen's cut rule:

$$\frac{\Gamma \rightarrow A \supset B \quad \Delta \rightarrow A}{\Gamma, \Delta \rightarrow B}$$

-the latter follows from the former by *Weakening*

-the former follows from the latter by the rule of *Contraction*:

$$\frac{A, A, \Gamma \rightarrow B}{A, \Gamma \rightarrow B} \text{Ctr}$$

The four axioms of implication are provable in *NDL*:

-Easy for axioms 1, 3, 4

- By a "hair's breadth" for axiom 2:

1. $A \supset (A \supset B) \rightarrow A \supset (A \supset B)$ initial sequent

2. $A \rightarrow A$ initial sequent

The four axioms of implication are provable in *NDL*:

-Easy for axioms 1, 3, 4

- By a "hair's breadth" for axiom 2:

1. $A \supset (A \supset B) \rightarrow A \supset (A \supset B)$ initial sequent
2. $A \rightarrow A$ initial sequent
3. $A \supset (A \supset B), A \rightarrow A \supset (A \supset B)$ from 1 by *Wk*
4. $A \supset (A \supset B), A \rightarrow A$ from 2 by *Wk*

The four axioms of implication are provable in *NDL*:

-Easy for axioms 1, 3, 4

- By a "hair's breadth" for axiom 2:

- | | |
|---|---------------------------|
| 1. $A \supset (A \supset B) \rightarrow A \supset (A \supset B)$ | initial sequent |
| 2. $A \rightarrow A$ | initial sequent |
| 3. $A \supset (A \supset B), A \rightarrow A \supset (A \supset B)$ | from 1 by <i>Wk</i> |
| 4. $A \supset (A \supset B), A \rightarrow A$ | from 2 by <i>Wk</i> |
| 5. $A \supset (A \supset B), A \rightarrow A \supset B$ | from 3 and 4 by <i>MP</i> |
| 6. $A \supset (A \supset B), A \rightarrow B$ | from 5 and 4 by <i>MP</i> |

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| 6. $A \supset (A \supset B), A \rightarrow B$ | from 5 and 4 by <i>MP</i> |
| 7. $A \supset (A \supset B) \rightarrow A \supset B$ | from 6 by <i>R$\supset$</i> |
| 8. $\rightarrow (A \supset (A \supset B)) \supset (A \supset B)$ | from 7 by <i>R$\supset$</i> |

Let's compare this to a derivation with independent context *MP*:

- | | |
|--|---------------------------------|
| 1. $A \supset (A \supset B) \rightarrow A \supset (A \supset B)$ | initial sequent |
| 2. $A \rightarrow A$ | initial sequent |
| 3. $A \supset (A \supset B), A \rightarrow A \supset B$ | from 1 and 2 by <i>MP-indep</i> |
| 4. $A \supset (A \supset B), A, A \rightarrow B$ | from 3 and 2 by <i>MP-indep</i> |

Let's compare this to a derivation with independent context *MP*:

1. $A \supset (A \supset B) \rightarrow A \supset (A \supset B)$ initial sequent
2. $A \rightarrow A$ initial sequent
3. $A \supset (A \supset B), A \rightarrow A \supset B$ from 1 and 2 by *MP-indep*
4. $A \supset (A \supset B), A, A \rightarrow B$ from 3 and 2 by *MP-indep*

The only way to proceed is to apply rule *Ctr*

Gödel's rule *MP* is in fact equivalent to shared-context *Cut*:

-Assume $R\supset$ has been applied:

1. $A, \Gamma \rightarrow B$ assumed
2. $\Gamma \rightarrow A \supset B$ from 1 by $R\supset$
3. $\Gamma \rightarrow A$ assumed
4. $\Gamma \rightarrow B$ from 2 and 3 by *MP*

Shared-context cut gives:

$$\frac{\Gamma \rightarrow A \quad A, \Gamma \rightarrow B}{\Gamma \rightarrow B}$$

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Shared-context cut gives:

$$\frac{\Gamma \rightarrow A \quad A, \Gamma \rightarrow B}{\Gamma \rightarrow B}$$

Gentzen invented cut elimination to have derivations with the *subformula property*.

Herbrand came very close to this in his formulation of a logical calculus without *MP*.

Gödel's *PAL*

From *RG*:

1. *Initial sequents*
2. *Weakening*
3. *Cut*

$$\frac{\Gamma \rightarrow A \quad A, \Delta \rightarrow B}{\Gamma, \Delta \rightarrow B} \textit{Cut}$$

4. *Right implication*

$$\frac{A, \Gamma \rightarrow B}{\Gamma \rightarrow A \supset B} R\supset$$

5. *Ground sequents*

$$A \supset B, A \rightarrow B$$

MP and *GS* are equivalent in the presence of *Cut*:

$$\frac{\Gamma \rightarrow A \quad \frac{\Gamma \rightarrow A \supset B \quad A \supset B, A \rightarrow B}{A, \Delta \rightarrow B}^{GS}}{\Gamma, \Gamma \rightarrow B}^{Cut}$$

1. $A \supset B \rightarrow A \supset B$
2. $A \rightarrow A$
3. $A \supset B, A \rightarrow B$

There is a further variant in one *Arbeitsheft*:

- take the *inverse* of $R_{\supset}$ as a rule
- apply it to $A \supset B \rightarrow A \supset B$
- result is *GS* $A \supset B, A \rightarrow B$

Gödel's method in *RG*:

- take the sequent to be derived, e.g., $p \supset (q \supset p)$ and all its subformulas
- form initial sequents with all the subformulas
- add instance of *GS* with all the subformulas
- apply *Wk*, *Cut*, and $R\supset$ to these, i.e.

Closure of initial and ground sequents under the three rules

Example: derivation of $p \supset (q \supset p)$

1. $p \rightarrow p$ initial sequent
2. $p, q \rightarrow p$ from 1 by Wk
3. $p \rightarrow q \supset p$ from 2 by $R\supset$
4. $\rightarrow p \supset (q \supset p)$ from 3 by $R\supset$

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2. $p, q \rightarrow p$ from 1 by Wk
3. $p \rightarrow q \supset p$ from 2 by $R\supset$
4. $\rightarrow p \supset (q \supset p)$ from 3 by $R\supset$

Second example: derivation of $(p \supset (p \supset q)) \supset (p \supset q)$

1. $p \supset (p \supset q), p \rightarrow p \supset q$ GS
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1. $p \supset (p \supset q), p \rightarrow p \supset q$ GS
2. $p \supset q, p \rightarrow q$ GS
3. $p \supset q, p \supset (p \supset q), p \rightarrow q$ from 2 by Wk
4. $p \supset (p \supset q), p \rightarrow q$ from 1 and 3 by Cut
5. $p \supset (p \supset q) \rightarrow p \supset q$ from 4 by $R\supset$
6. $\rightarrow (p \supset (p \supset q)) \supset (p \supset q)$ from 5 by $R\supset$

It is seen that the *Cut* on line 4 is one with shared contexts

Example: derivation of $p \supset (q \supset p)$

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2. $p, q \rightarrow p$ from 1 by Wk
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Second example: derivation of $(p \supset (p \supset q)) \supset (p \supset q)$

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2. $p \supset q, p \rightarrow q$ GS
3. $p \supset q, p \supset (p \supset q), p \rightarrow q$ from 2 by Wk
4. $p \supset (p \supset q), p \rightarrow q$ from 1 and 3 by Cut
5. $p \supset (p \supset q) \rightarrow p \supset q$ from 4 by $R\supset$
6. $\rightarrow (p \supset (p \supset q)) \supset (p \supset q)$ from 5 by $R\supset$

It is seen that the *Cut* on line 4 is one with shared contexts

What is the justification of the restriction to subformulas?

Derivations with Cut need not have the subformula property!

Cut elimination for *PAL*

1. Take uppermost cut
2. Initial sequents and weakening handled the usual way
3. If right premiss derived by $R\supset$, permute as in

$$\frac{\Gamma \rightarrow C \quad \frac{C, A, \Delta \rightarrow B}{C, \Delta \rightarrow A \supset B} R\supset}{\Gamma, \Delta \rightarrow A \supset B} Cut \quad \sim \quad \frac{\Gamma \rightarrow C \quad \frac{C, A, \Delta \rightarrow B}{A, \Gamma, \Delta \rightarrow B} Cut}{\Gamma, \Delta \rightarrow A \supset B} R\supset$$

Repeat until right premiss an instance of *GS*

3.1. Left premiss derived by $R\supset$, two cases:

3.1.1.

$$\frac{\frac{A, \Gamma \rightarrow B}{\Gamma \rightarrow A \supset B} R\supset \quad A \supset B, A \rightarrow B}{A, \Gamma \rightarrow B} Cut$$

3.1.2.

$$\frac{\frac{A, \Gamma \rightarrow B}{\Gamma \rightarrow A \supset B} R\supset \quad (A \supset B) \supset C, A \supset B \rightarrow C}{(A \supset B) \supset C, \Gamma \rightarrow C} Cut$$

3.2. Left premiss an instance of *GS*, again two cases:

3.2.1

$$\frac{A \supset B, A \rightarrow B \quad B \supset C, B \rightarrow C}{A \supset B, B \supset C \rightarrow C} \textit{Cut}$$

3.2.2.

$$\frac{A \supset (B \supset C), A \rightarrow B \supset C \quad B \supset C, B \rightarrow C}{A \supset (B \supset C), A, B \rightarrow C} \textit{Cut}$$

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$$\frac{A \supset (B \supset C), A \rightarrow B \supset C \quad B \supset C, B \rightarrow C}{A \supset (B \supset C), A, B \rightarrow C} \textit{Cut}$$

Cuts remain in derivations so we have to consider a cut that hits an uppermost one.

This can be done but I won't give the details here.

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With independent contexts, *Ctr* must enter as an explicit rule, with the problem:

$$\frac{\Gamma \rightarrow A \quad \frac{A, A, \Delta \rightarrow C}{A, \Gamma, \Delta \rightarrow C} Ctr}{\Gamma, \Delta \rightarrow C} Cut$$

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The transformation would be:

$$\frac{\Gamma \rightarrow A \quad \frac{\Gamma \rightarrow A \quad A, A, \Delta \rightarrow C}{A, \Delta \rightarrow C} Cut}{\frac{\Gamma, \Gamma, \Delta \rightarrow C}{\Gamma, \Delta \rightarrow C} Ctr^*} Cut$$

The size of derivation of the right premiss of the last cut has grown.

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The size of derivation of the right premiss of the last cut has grown.

Note that derivations of the premisses of an uppermost cut are linear, so if there is there a duplication it is even in a top sequent.

Conclusions:

There is no trace of Gödel having understood the significance of cut elimination.

No use of the subformula property. It states for implicative logic:

If a formula is derivable, it is derivable by the logical rules of implication and the structural rules of weakening and contraction.

Gödel's result about the decidability of implicative logic follows at once from Gentzen's proof of the decidability of intuitionistic propositional logic, a corollary to cut elimination.

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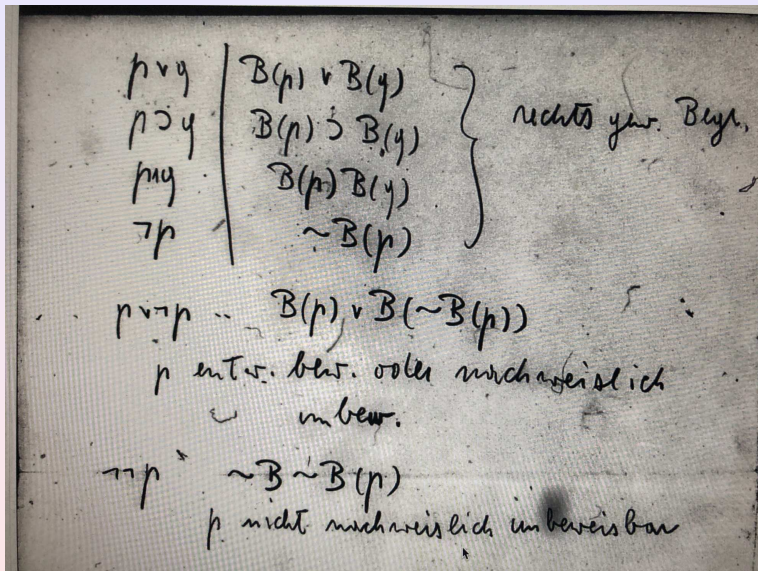
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All methods would likely lead to an understanding of Gödel's own proof, given in terms of old-fashioned axiomatic logic few understand today.

2. The modal interpretation of intuitionistic logic

Gödel's 1933 translation



Gödel's axioms for provability in the trial lecture are:

1. $B(p) \supset p$
2. $B(p \supset q) \cdot B(p) \supset B(q)$
3. $B(p) \supset BB(p)$

Table. Axioms of provability

The rule of inference is that if p has been proved, $B(p)$ can be inferred. Today Gödel's system is known as the modal logic **S4**.

The trial lecture shows how Gödel proved his 1933 result:
Theorem. *The modal translations of Heyting's axioms are
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Outline of proof: Instantiate the axioms with $B(p), B(q), \dots$ in place of the propositional variables $p, q, \dots$. Then apply the derivable rule by which from $p \supset q$ (assumed provable), $B(p) \supset B(q)$ can be inferred.

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Gödel conjectured in 1933 that the correspondence is exact in both directions:

*A is provable in intuitionistic logic if and only if its modal translation $U(A)$ is provable in **S4**.*

The *Results on Foundations*, 1941, contains a 25-page syntactic proof of this conjecture.

2) $\sim \vdash A: \mathcal{L} \text{ per } \sim$

$$M[A_1 \dots A_n]' \equiv_H M'[A_1' \dots A_n']$$

$< M' = \text{Hyp. Ident.} \hookrightarrow M - \text{ng}$

(U Th 62)

13 Hyp $\vdash A, B \in \text{Lerns}$

$$(\mathcal{L}(A) \supset \mathcal{L}\mathcal{L}(A))' \text{ H. 10}$$

$$(\mathcal{L}(A), \mathcal{L}(A \supset B) \supset \mathcal{L}(B))' \text{ H. 10}$$

Bew 1. $\vdash A' \equiv_H \mathcal{L}(A)' \supset \mathcal{L}\mathcal{L}(A)' = A' \supset A'$

2. $\vdash \equiv_H A'(A \supset B)' \supset B'$ ab $\checkmark$ U Hyp 10

14 Hyp $\hookrightarrow A' \supset (A \supset B)' \text{ H. 10} \supset \checkmark B'$

U. Hyp 10

15 Hyp $\hookrightarrow A' \supset \text{H. 10} \supset \checkmark \mathcal{L}(A)'$

$$\hookrightarrow \mathcal{L}(A)' = A'$$

Th 63

$\hookrightarrow A - \text{H. 2 Lerns. 100 p.}$
 $\supset A' \supset \text{H. 100}$

Bew

U Hyp 11, 12, 13 $\supset \checkmark \mathcal{L}(A)$
 $\supset \text{Lerns. 11} \supset \checkmark p$

U Hyp 14, 15 $\hookrightarrow \checkmark \text{ per } \checkmark$

Th 64

$\hookrightarrow A - p \supset \text{Hyp. 100} \supset U(A)$

[vgl. Hyp 9] $\supset \checkmark \text{ Lerns. 100} \supset A \supset \text{Hyp.}$

Bew

U Th 63 $\supset U(A)' \supset \text{Hyp. 100}$

$$< U(A)' \equiv_H A \supset \text{Hyp. 9}^*$$

$\mathcal{L}_\sim$ Modalitäten im Lernskalkül (Vgl. $\mathcal{L}(A) \supset \mathcal{L}\mathcal{L}(A)$)

Schemen an $\sim$ $\supset \checkmark \mathcal{L}$ an $\sim$ Modalitäten

$\mathcal{L}$ in $\sim$ einfach $\hookrightarrow$

$\supset \checkmark \mathcal{L}$ Modalitäten



* Forts. p. 249.

Idea of proof for Thm 63: Translate any proof of A in **S4** into a proof of A' in H.

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The proof turned out to use the *propositional case of Barr's theorem*:

Each formula A is classically equivalent to a conjunction $A_1 \& \dots \& A_k$ with

$$A_i = \neg P_1 \vee \dots \vee \neg P_m \vee Q_1 \vee \dots \vee Q_n$$

Let

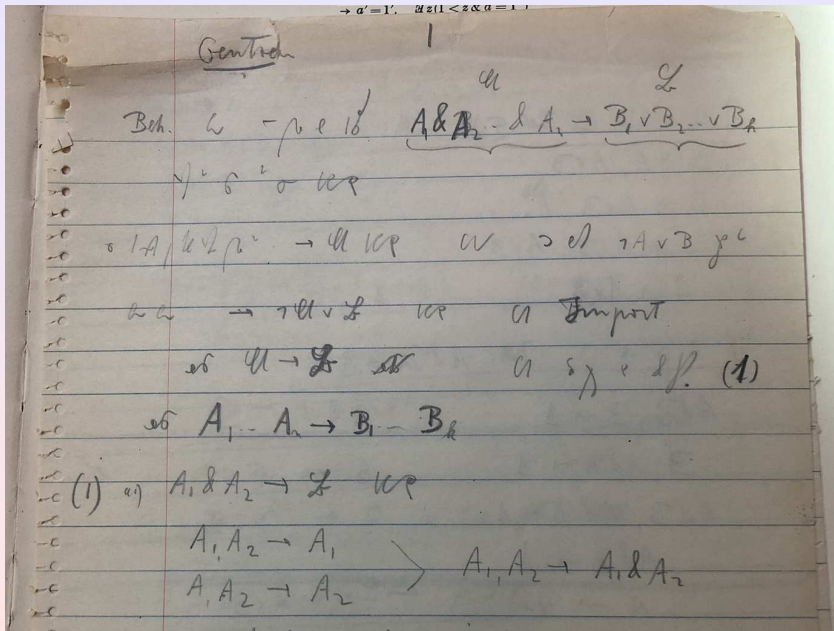
$$B_i = P_1 \& \dots \& \neg P_m \supset Q_1 \vee \dots \vee Q_n$$

Then B_i classically equivalent to A_i , **but**:

A classically derivable if and only if $B_1 \& \dots \& B_k$ intuitionistically derivable.

Question: Where did the propositional Barr's theorem come from?

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3. Gödel's "constructively strongest" formula

From Gödel's *Princeton Lectures on Intuitionism*, spring 1941:

Take a proposition of the form $\forall x A(x)$. Then the statement $\exists x \neg A(x)$ evidently also is a kind of negation of A but stronger than $\neg \forall x A(x)$ because $\exists x \neg A(x) \rightarrow \neg \forall x A(x)$ can be asserted but not vice versa.

[[Remark about translation into Heyting calculus]]³⁰

Remark. Let $\bar{A}$ designate the formula that arises from A through driving the $\neg$ sign inwards as much as possible (the basic relations assumed decidable):

Then $\bar{A} \rightarrow A$ is provable in the Heyting calculus.

Proof. By induction on the number of logical signs in A , where however, if A begins with $\neg$, two steps are to be taken, i.e., we have $A = \neg(B \vee C)$ or $\neg(B \& C)$

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etc and the theorem holds already for B, C .

Main theorem, Gentzen's proof of consistency³¹

Theorem 57. Let $\mathfrak{A}$ be an expression (without free variables)³² and $\mathfrak{Q}$ a subexpression of $\mathfrak{A}$,³³ and more precisely,

$$\mathfrak{Q} = ((x)\Phi) \vee \mathfrak{P}$$

Let further

³⁰ [[Gödel's title from the summary: $\bar{A} \rightarrow A$ provable in Heyting calculus if $\bar{A}$ is the "most constructive" formulation of A]]

³¹ [[Gödel's title from the summary: Main lemma of the proof of consistency of number theory through choice sequences]]

³² This means that the formula has at least two different quantifications designated

Conversion steps to negation normal form

$$1. P \rightsquigarrow P$$

$$2. \neg P \rightsquigarrow \neg P$$

$$3. \neg(A \& B) \rightsquigarrow \neg A \vee \neg B$$

$$4. \neg(A \vee B) \rightsquigarrow \neg A \& \neg B$$

$$5. \neg(A \supset B) \rightsquigarrow A \& \neg B$$

$$6. \neg\neg A \rightsquigarrow A$$

$$7. \neg\forall x A(x) \rightsquigarrow \exists x \neg A(x)$$

$$8. \neg\exists x A(x) \rightsquigarrow \forall x \neg A(x)$$

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4. $\neg(A \vee B) \rightsquigarrow \neg A \& \neg B$
5. $\neg(A \supset B) \rightsquigarrow A \& \neg B$
6. $\neg\neg A \rightsquigarrow A$
7. $\neg\forall x A(x) \rightsquigarrow \exists x \neg A(x)$
8. $\neg\exists x A(x) \rightsquigarrow \forall x \neg A(x)$

The uniquely determined formula obtained from a formula A in this manner I denote by $\overline{A}$. Then $\overline{\overline{A}} \equiv A$ in classical logic and in intuitionistic logic we have $\overline{A} \rightarrow A$ can be asserted (but not vice versa). So $\overline{A}$ is the intuitionistically strongest formula which is classically equivalent to A under these rules of shifting negation. **I don't prove this.**

Gödel's next point in the *Princeton Lectures* is (ibid.):

Now constructive negation of course defined by $\sim A = \overline{\neg A}$
and now the second theorem connecting classical and
intuitionistic logic reads like this: If A is classically provable
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then $\neg \sim A$ is intuitionistically demonstrable.

Theorem 1. $\vdash_I NNF(A) \supset A$.

Theorem 2. Soundness of Gödel's 1941 translation from classical to intuitionistic logic. *If $\vdash_K A$, then $\vdash_I \neg \sim A$.*

Proof. $\sim A$ is by definition $NNF(\neg A)$ and by Theorem 1, $\vdash_I NNF(\neg A) \supset \neg A$, so by contraposition, $\vdash_I \neg \neg A \supset \neg NNF(\neg A)$. It remains to note that $\vdash_I A \supset \neg \neg A$. QED.

Definition 3. Constructively strongest formula. *B is the constructively strongest formula classically equivalent to a given formula A if, whenever C is classically equivalent to A and $\vdash_I C \supset A$, then $\vdash_I B \supset C$.*

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Here is a counterexample to Gödel's claim, with $C = A = \neg P \vee Q$:

$P \supset Q$ is in negation normal form, classically equivalent to $\neg P \vee Q$, but

$$\not\vdash_I (P \supset Q) \supset \neg P \vee Q$$

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Here is a counterexample with no implications.

$$\vdash_K P \equiv P \& (Q \vee \neg Q) \text{ and } NNF(P) = P$$

but

$$\not\vdash_I NNF(P) \supset P \& (Q \vee \neg Q)$$

The solution:

Theorem 5. Gödel's truth table theorem. *If A is a tautology and $\pm P_1 \& \cdots \pm P_n$ any conjunction of the atoms and negations of atoms in A , then $\vdash_I \pm P_1 \& \cdots \pm P_n \supset A$.*

The solution:

Theorem 5. Gödel's truth table theorem. *If A is a tautology and $\pm P_1 \& \dots \pm P_n$ any conjunction of the atoms and negations of atoms in A , then $\vdash_I \pm P_1 \& \dots \pm P_n \supset A$.*

The easiest way to prove this result is to show that if A is a tautology in the atoms $P_1, \dots, P_n$, its classical derivation can be converted into an intuitionistic derivation of

$$(P_1 \vee \neg P_1) \& \dots \& (P_n \vee \neg P_n) \supset A$$

The antecedent is equivalent to the disjunction of all the **primitive conjunctions** (Gödel's term) $\pm P_1 \& \dots \pm P_n$.

Theorem 5 in Gödel's own words reads (p. 50):

We have the remarkable fact that all formulas corresponding to the truth tables of any of the logical operations $\neg, \rightarrow, \cdot, \vee$, are identities also in intuitionistic logic.

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If A is not a tautology, let $C_1, \dots, C_k$ be the primitive conjunctions $\pm P_1 \& \dots \pm P_n$ of the lines on which A obtains the value true.

Then $\vdash_I C_i \supset A$ for each i (with $\perp \supset A$ if A is a contradiction) by which we get:

$$\vdash_I C_1 \vee \dots \vee C_k \supset A$$

The contrary implication holds only classically, and we have

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The left side is the **disjunctive normal form** $DNF(A)$ of A .

Theorem 6. Constructively strongest formula. *$DNF(A)$ is the constructively strongest formula classically equivalent to A .*

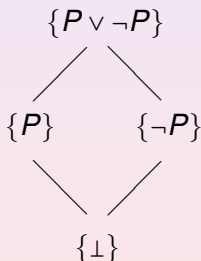
By above, we have $\vdash_I DNF(A) \supset A$. Strongest is shown below.

For formulas in one atom P , the constructively strongest formulas are

$$\perp, P, \neg P, P \vee \neg P$$

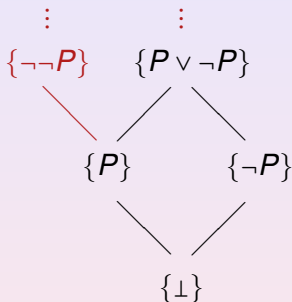
well known from the Rieger-Nishimura lattice.

The classical Boolean implicative lattice in one atom:



The intuitionistic Heyting lattice in one atom:

classical tautologies



Just one non-tautological formula $\neg\neg P$ added.

The rest an infinity of intuitionistically inequivalent classical tautologies.

With two atoms P and Q , there are four primitive conjunctions:

$$P \& Q \quad P \& \neg Q \quad \neg P \& Q \quad \neg P \& \neg Q$$

There are, as with one atom, the two non-tautological formulas $\neg\neg P$ and $\neg\neg Q$, and an infinity of intuitionistically inequivalent formulas in P and in Q that are classical tautologies.

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Question: Are there, beyond the formulas $\neg\neg P$ and $\neg\neg Q$, other non-tautological intuitionistically inequivalent formulas, and if so, how many are they?

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Question: Are there, beyond the formulas $\neg\neg P$ and $\neg\neg Q$, other non-tautological intuitionistically inequivalent formulas, and if so, how many are they?

Guess 1: A few.

Guess 2: Many.

Guess 3: A huge number.

Guess 4: An infinity.

The classical Boolean implicational lattice in two atoms:

$$(P \& Q) \vee (P \& \neg Q) \vee (\neg P \& Q) \vee (\neg P \& \neg Q)$$

Top level

$$(P \& Q) \vee (P \& \neg Q) \vee (\neg P \& Q) \quad (P \& Q) \vee (P \& \neg Q) \vee (\neg P \& \neg Q)$$

$$(P \& Q) \vee (\neg P \& Q) \vee (\neg P \& \neg Q) \quad (P \& \neg Q) \vee (\neg P \& Q) \vee (\neg P \& \neg Q)$$

Implication level 2

$$(P \& Q) \vee (P \& \neg Q) \quad (P \& Q) \vee (\neg P \& Q) \quad (P \& Q) \vee (\neg P \& \neg Q)$$

$$(P \& \neg Q) \vee (\neg P \& Q) \quad (P \& \neg Q) \vee (\neg P \& \neg Q) \quad (\neg P \& Q) \vee (\neg P \& \neg Q)$$

Implication level 1

$$P \& Q \quad P \& \neg Q \quad \neg P \& Q \quad \neg P \& \neg Q$$

Bottom level

$\perp$

There are 32 relations of *immediate implication* between the DNF's.

Theorem 7. *Intuitionistic logic adds no formulas inside the classical Boolean lattice. This holds for any number of atomic formulas.*

Proof idea. For $P \& Q$, assume C to be a non-contradictory formula such that $\vdash_I C \supset P \& Q$. By Theorem 6, $\vdash_I DNF(C) \supset P \& Q$. Prove by induction on the length of C that $\vdash_I C \equiv P \& Q$.

The proof generalizes easily to any number of atomic formulas.

Now back to the quiz:

Proposition. (Answer 4 to the quiz.) *From two atomic formulas on, there is an infinity of intuitionistically inequivalent formulas in propositional logic that are not classical tautologies.*

Sources and Studies in the History of Mathematics
and Physical Sciences

Maria Hämeen-Anttila
Jan von Plato *Editors*

Kurt Gödel

The Princeton Lectures on Intuitionism

 Springer

Overall conclusions and challenges:

-Machine reading of shorthand. If this succeeds, thousands of lecture notes from the past will become accessible, opening a new chapter in the history of science: How were the new theories taught to students?

-With Gödel in particular, and beyond the reading task: to understand his proofs

Greatest challenges in the latter:

1. Understanding his formal system of intuitionistic analysis

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Items 2 and 4 are the greatest open questions in the foundations of mathematics

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